Emotion Discovery and Reasoning its Flip in Conversation(EDiReF)

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Index

S. No.	Contents	Page No.
1.	Abstract	1
2.	Introduction - Literature Review - Research Gap - Research Objectives	1
3.	Proposed Architecture	4
4.	Programming Environment/Tools Used	5
5.	References	5

Emotion Discovery and Reasoning its Flip in Conversation(EDiReF)

Abstract

Emotions have always been an inseparable part of human communication, ranging from face-to-face conversations, speeches involving facial expressions, and even text-based conversations, which turn out to be one of the most prevalent mediums of passing information in the current scenario. They are always one of the major driving forces of the course of conversation, its outcomes, and consequences on the involved parties as well as the associated environment. This prime importance of emotions brings with them a crucial challenge of recognizing emotions accurately in a sequence of dialogues. The subtleties of emotion variation are dynamic in nature, due to the possibilities of affirmation, negation, sarcasm, and other related communication components existing in the set of dialogues. Moreover, the emotional tendencies of the involved speakers play an important role in determining the conversation results. Thus, in accordance with the same, the domain of emotion recognition and figuring out the triggers of emotion flip have found emergence in recent times and gained attention from researchers for developing mechanisms to accomplish the related objectives. This research work will be oriented towards the tasks of emotion recognition in conversations (ERC) and emotion flip reasoning (EFR).* In this research study, we aim to devise a meticulous mechanism for recognizing emotions in a series of conversations. The system will be implemented in order to understand the speaker-oriented information, and the dynamics including affirmation phrases, sarcasm-inclusive phrases, and important features affecting the emotion-based situation of the conversation. In addition to this, we propose a mechanism for detecting and extracting the triggers existing in the course of dialogues which lead to a significant change of emotional state of the involved parties. The system will incorporate transliteration models, vectorization models for converting dialogues into word embeddings, and large language models (LLMs) to understand the context as well as the sentiments of the dialogues, eventually followed by classifiers and EFR methodologies for acquiring the desired outputs.

Keywords: Emotion Recognition in conversations (ERC), Emotion Flip Recognition (EFR), Large Language Models, Sentiment Analysis, context-cues, speaker-aware mechanism

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*This research work will be a part of the Shared Task-10 at the 18th International Workshop on Semantic Evaluation (SemEval 2024)

1 Introduction

Human communication is significantly shaped by emotions, which have an impact on the manner, purpose, and results of conversations. Although emotions have a big impact on communication dynamics, it's still difficult to understand how they change in the moment. Understanding the subtleties of emotional swings during conversations can be difficult due to the dynamic nature of emotions. Speakers in a conversational dialogue may be in a range of emotional states, and their interactions are crucial to comprehending the emotional discourse of the dialogue. Understanding the speaker-specific emotion changes during a discussion is more important than merely detecting the emotions. We need to understand the explanation or cause of the speaker's emotional swings in order to specifically comprehend the dynamics of emotion in a discourse.

1.1 Literature Review

Recognizing emotions and detecting the triggers for change of emotions in the course of a dialogue narrative or any other means of conversation have lately seen a phase of prime importance in the field of Artificial Intelligence and Natural Language Processing, in particular. This has been primarily due to the high dependence of various research areas on this specialism, ranging from virtual assistants, business survey analysis, market research, law enforcement surveillance, psychological diagnosis and treatment, smart education, etc.

To combat this highly emerging subdomain, several research works have been carried out dealing with the art of detecting various emotions through text and other forms of data, as illustrated in Table 1. For instance, in the research study [1]), an attempt has been made to address the challenges associated with the task of emotion recognition in conversations along with the existing approaches and their drawbacks for doing the same.

In [6], a combination of semantic/syntactic and emotional information has been incorporated by adopting pre-trained word representations to recognize emotions from textual data. Another research work [2], involves the task of emotion classification of web blog corpora using support vector machine (SVM) and conditional random field (CRF) machine learning techniques.

In [7], a comparative analysis was performed among BERT, RoBERTa, DistilBERT, and XLNet for text-based emotion recognition. An accuracy of 0.7431, 0.7299, 0.7009, and 0.6693 was achieved for RoBERTa, XLNet, BERT, and DistilBERT (respectively). In the research study [4], a novel approach was proposed with EmoTxt, a toolkit for emotion recognition from text and for the training of custom emotion classification models.

Another research study [5], proposed a multimodal emotion detection framework that extracts multimodal features from conversational videos and hierarchically models the self- and inter-speaker emotional influences into global memories. In a similar study [9], a context- and sentiment-aware framework, termed Sentic GAT, which tends to select the commonsense knowledge consistent with the context semantics and sentiment of the target utterance, was proposed to solve this challenge.

References	Paper Title	Dataset Used	Models Used	Results Obtained
[2]	Emotion Classification Using Web Blog Corpora	Blog Posts	SVM and CRF	Refer Table 7 (in publication)
[3]	Multiple Emotions Detection in Conversation Transcripts	Cornell Movie Dialog dataset, ISEAR data	Neural Network	Refer Table 4 (in publication)
[4]	EmoTxt: A Toolkit for Emotion Recognition from Text	Stack Overflow and Jira posts	EmoTxt classification models	Refer Table 2 (in publication)
[5]	ICON: Interactive Conversational Memory Network for Multimodal Emotion Detection	IEMOCAP and SEMAINE	Interactive Conversational memory Network (ICON)	Refer Table 6 (in publication)
[6]	Semantic-Emotion Neural Network for Emotion Recognition From Text	Aggregated annotated corpora for Emotion Classification in Text	Semantic-Emotion Neural Network (biLSTM & CNN)	Refer Table 3 (in publication)
[7]	Comparative Analyses of BERT, RoBERTa, DistilBERT, and XLNet for Text-based Emotion Recognition	ISEAR data	BERT, RoBERTa, DistilBERT, and XLNet pre-trained transformer models	Refer Table 2 (in publication)
[8]	Contextualized Emotion Recognition in Conversation as Sequence Tagging	IEMOCAP, MELD and Dailydialogue	DialogueRNN	Refer Table 4 (in publication)
[9]	Context- and Sentiment-Aware Networks for Emotion Recognition in Conversation	DailyDialog, MELD and EmoryNLP	Sentic GAT	Refer Table 3 (in publication)

Table 1 Existing Work for Emotion Recognition

In the research work [3], a method of predicting emotions from multi-label conversation transcripts has been developed. The proposed methodology includes building an emotion lexicon bootstrapped from Wordnet following the notion of Plutchik's basic emotions and dyads. A similar study [8], proposes a method to model the ERC task as sequence tagging where a Conditional Random Field (CRF) layer is leveraged to learn the emotional consistency in the conversation. With this, LSTM-based encoders are employed that capture self and inter-speaker dependency of interlocutors to generate contextualized utterance representations which are fed into the CRF layer. For capturing long-range global context, a multi-layer Transformer encoder has been used to enhance the LSTM-based encoder.

Several research works have been carried out for the task of reasoning of emotion flip (EFR) as well, as mentioned in Table 2. For instance, in [10], the task of Instigator-based Emotion Flip Reasoning (EFR) has been kept forward, which aims to identify the instigator behind a speaker's emotion flip within a conversation. As a part of this work, a novel neural architecture called TGIF has been developed, which leverages Transformer encoders and stacked GRUs to capture the dialogue context, speaker dynamics, and emotion sequence in a conversation.

References	**Paper Title**	**Dataset Used**	**Models Used**	**Results Obtained**
[11]	Emotion Estimation and Reasoning Based on Affective Textual Interaction	WordNet-Affect Database of ITS-irst	Emotion estimation module based on Word-spotting technique	-
[12]	Recognizing Emotion Cause in Conversations	RECCON (built in this work using IEMOCAP and DailyDialog)	RoBERTa Base and Large, SpanBERT Fine-tuned on SQuAD, ECPE-2D, ECPE-MLL, RankCP	Refer Tables 5 and 6 (in publication)
[10]	Emotion Flip Reasoning in Multiparty Conversations	MELD-I	TGIF	Refer Table 4 (in publication)

Table 2 Existing Work for Emotion Flip Recognition

In the research study [11], the methodology of emotion estimation has been proposed that assesses the affective content from textual messages. to detect emotion from chat or other dialogue messages and to employ animated agents capable of emotional reasoning based on textual interaction.

Another research study [12], featured a new annotated dataset for the task of recognizing emotion cause in conversations, RECCON. This study formulated two distinct subtasks of recognizing emotion cause in conversations: (i) causal span extraction and

(ii) causal emotion entailment. For the first subtask, RoBERTa Base, and SpanBERT Fine-tuned on SQuAD, were incorporated in this study, while the second subtask involved the use of RoBERTa Base and Large, ECPE-2D, ECPE-MLL, and RankCP.

1.2 Research Gap

- Lexicon-based and modern deep learning-based, vanilla emotion recognition ignores conversation-specific factors such as the presence of contextual cues, the temporality in speakers' turns, or speaker-specific information.
- The state-of-the-art methods keep mimicking the same emotion for a particular party and fail in most cases where a change occurs.
- If two speakers use completely opposite emotions to express their common interest or opinion, it is not possible for a vanilla emotion recognizer to understand the positive emotion of both the interlocutors on the aspect of the opinion.
- Naturally, emotion recognition in multiparty conversations is more challenging in comparison with dyadic conversations due to the difficulty in tracking individual speaker states and handling co-references.
- An ERC system, incapable of detecting sarcasm, mostly fails to predict the emotion of the sarcastic utterances correctly.

1.3 Research Objectives

- To devise a mechanism for interpreting conversation-specific factors, by comprehending contextual cues, and understanding the temporality in speakers' turns, along with the speaker-specific information.
- To implement a system capable of monitoring the emotion sequence, tracking the triggers which can possibly result in a turn of speaker's emotions.
- To ensure that the proposed system is able to comprehend the assertion and negation aspects resulting out a dialogue.
- To handle multiparty conversations in addition to dyadic conversations, while tracking individual speaker states and handling co-references.
- To prepare a methodology for handling sarcasm-inclusive conversations accurately.

2 Proposed Architecture

In this paper, we propose a framework based on six different sets of models, as depicted in Figure 1. One of the models will be used for recognizing emotions in the conversation sequence (ERC) while the other one will be implemented for determining the emotion-flip triggers (EFR). The remaining models are for data preprocessing, deriving context, and sentiment analysis in the conversation.

The potential datasets which can be used for this research work are listed below:

- Dataset provided by the shared task organizers containing dyadic and multiparty conversations
- IEMOCAP
- MELD etc

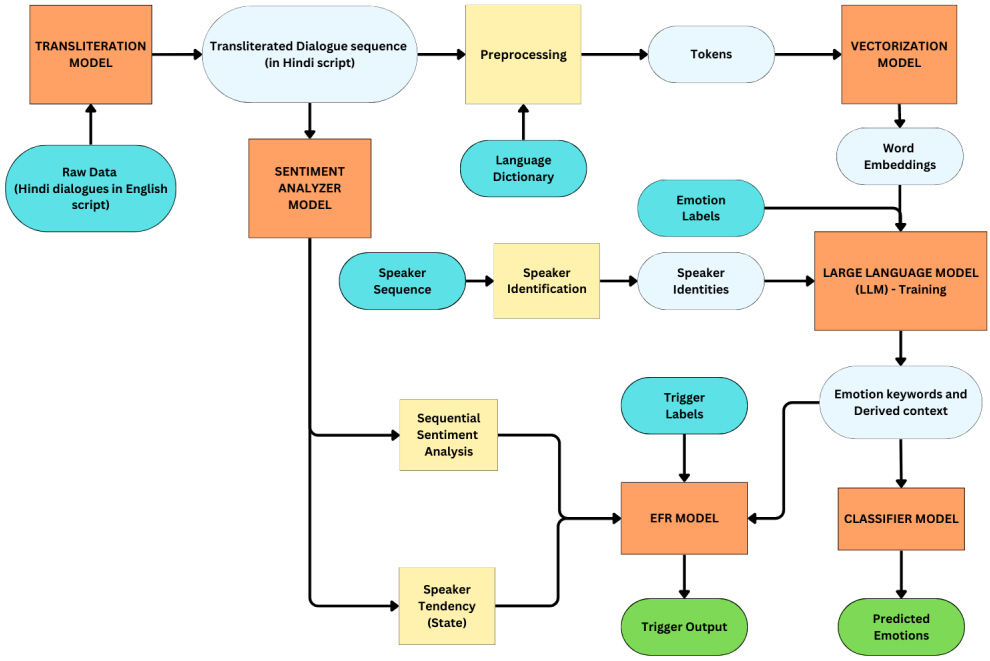

Fig. 1 Proposed Architecture Pipeline

The proposed architecture is as follows:

- The raw data (in English script) will be passed on to the Transliteration model which will generate a Transliterated dialogue sequence (in Hindi script).
- The obtained which will be preprocessed to form tokens as input for the Vectorization model, in order to produce word embeddings for the same.
- Alongside, the speaker sequence will be analyzed to comprehend the individual speaker identities corresponding to the dialogues.
- Eventually, the word embeddings for the dialogues, along with the emotion labels and speaker identities will be fed to a Large Language Model for training, thereby producing emotion keywords and comprehending the context for the utterances.
- Based on these derivations, the classifier will predict the emotions for the respective utterances producing results for Emotion Recognition in Conversation (ERC). Meanwhile, the EFR-oriented mechanism will process the sequential sentiment analysis from the sentiment analyzer in addition to the speaker states and context derivations to determine the trigger points which are responsible for flip of emotions (EFR).

3 Programming Environment/Tools Used

1. **System Environment:** Python, Google Colab/Jupyter Notebook, GitHub
2. **Frameworks/Libraries:** PyTorch, NLTK etc.

3. **Models:** Transliteration Models, Vectorization Models, Large Language Models (LLMs), Classifiers, EFR Models etc.

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